

Hardware Engineering Technical Report

Team A3

June 25, 2026

Team Members

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1 Introduction

This report documents the complete hardware design project realised by Team A3 during the Hardware Engineering Lab, Summer Term 2026. Together, the two parts demonstrate the full embedded-hardware development cycle: *design and verify in simulation* → *prototype on FPGA dev-board* → *tape-out to custom PCB*. The project is composed of two tightly coupled parts:

1. **VHDL soft-core CPU on Nexys A7 (Vivado)** — an 8-bit register-based CPU with a multi-operation ALU, implemented in VHDL and synthesised for an Artix-7 FPGA using Xilinx Vivado. The system supports addition, subtraction, multiplication, and division, with results displayed in signed decimal on a seven-segment display.
2. **Custom PCB design (Altium)** — a four-layer printed circuit board built around a Lattice iCE40UP5K-SG48I FPGA, designed to take the prototyped logic from the development board to a real, manufacturable hardware platform.

2 Concept Description

2.1 Main application

The prototype is an **interactive 8-bit calculator** with a four-register file. The user loads signed 8-bit values into registers R0–R3 and executes arithmetic operations (ADD, SUB, MUL, DIV) between any two registers, storing the result in a destination register. Results are displayed in signed decimal on the seven-segment display. Division results are shown as *quotient remainder* using the decimal point, and error conditions (overflow, divide-by-zero) display 0F.

2.2 Components and interfaces

Component	Description
Nexys A7 board	Artix-7 FPGA development board; hosts the entire VHDL design
SW[8:0]	9 slide-switches: mode, sign, magnitude, register select, operation
btn_exec	Edge-triggered execute button — loads a value or launches the ALU
btn_rst	Asynchronous reset; clears all registers and CPU state
7-segment display	8-digit multiplexed display; shows signed result and remainder
LEDs [7:0]	Mirror SW[7:0] for live switch-state debugging
Lattice iCE40UP5K	FPGA on the custom PCB (Part 2 of the project)
AMS1117 (×2)	LDO regulators — 3.3 V and 1.2 V supply rails on the PCB
Winbond W25Q16	16 Mb SPI NOR Flash — stores the FPGA configuration bitstream

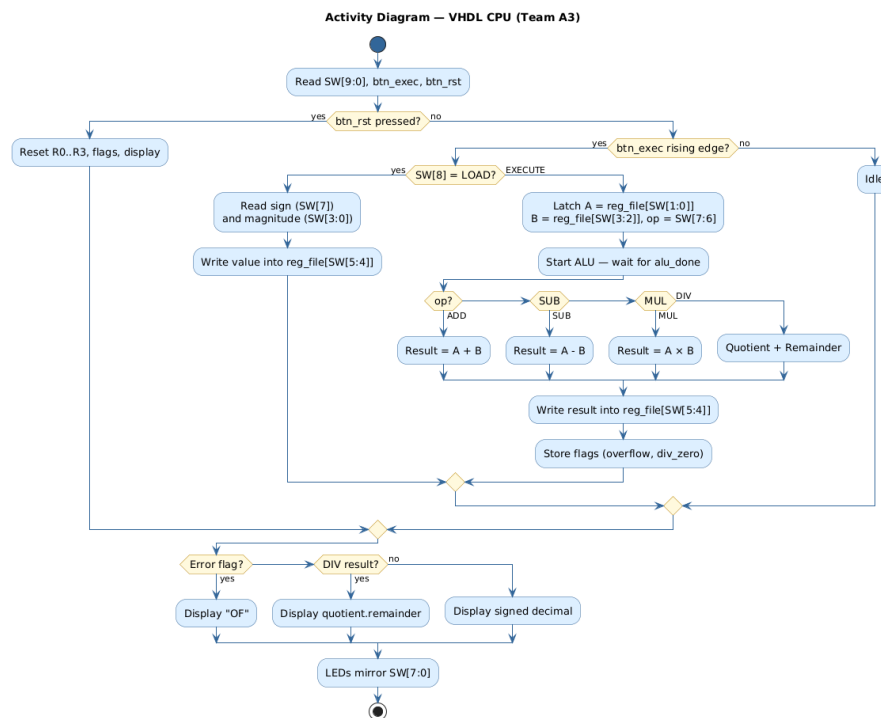


Figure 1: Activity diagram — VHDL CPU system

3 Project / Team Management

3.1 Development workflow

The team followed a two-phase hardware development flow:

1. **VHDL phase** — architecture definition, RTL coding in VHDL, functional simulation in Vivado, synthesis and implementation for Artix-7, on-board testing with Nexys A7.
2. **PCB phase** — schematic capture in Altium Designer, four-layer layout, DRC verification, and Gerber export to JLCPCB production standards.

3.2 Project method

The team adopted a **sequential milestone-based approach**. Within each phase the team applied iterative peer review, a verification-first policy (no step advanced without passing its check), and Git-based version control for all source files.

3.3 Task distribution

Task	Details	Team
CPU architecture	Top-level entity, register file, mode/switch decode	All
ALU design	ADD, SUB, MUL, DIV; FSM; carry/overflow/div-zero flags	All
Display subsystem	BCD conversion, 7-seg decoder, multiplexed refresh	All
Vivado flow	Simulation, synthesis, implementation, bitstream	All
PCB schematic	FPGA, power supply, SPI Flash, UI peripherals	All
PCB layout & route	4-layer stack-up, autoroute, serpentine tuning	All
DRC & compliance	Via rules, JLCPCB verification, Gerber export	All
Documentation	Report writing, diagrams, figures	All

4 Technologies

4.1 VHDL and Vivado

The CPU is described entirely in **VHDL** (VHSIC Hardware Description Language) at the register-transfer level (RTL). Xilinx **Vivado 2024.x** was used for synthesis, implementation, and bitstream generation targeting the **Artix-7 XC7A100T** on the Nexys A7 board. Functional verification was performed with Vivado's built-in simulator before programming the board.

4.2 Hardware description: `cpu_top` architecture

The top-level entity `cpu_top`, has 4bit input but result accumulate to 8bit registers and outputs. The CPU instantiates three sub-components and contains the complete datapath and control logic:

- **alu** — arithmetic logic unit with a clocked FSM (described in Section 5).
- **binary_to_bcd** — double-dabble converter producing hundreds, tens, and ones BCD digits from an 8-bit binary value.
- **seven_segment_decoder** — combinational 4-to-7 segment decoder; instantiated four times (ones, tens, hundreds, remainder-ones).

The datapath is controlled by a register file of four 8-bit registers (R0–R3), which serve as both operand sources and result destinations. In load mode, a signed value is formed using the `twos_comp8` function and written directly into the selected register. In execute mode, the selected operands are latched into the ALU, the operation is performed over one or more clock cycles, and the result is written back through the FSM writeback stage. All arithmetic relies on the `add8_ci` ripple-carry adder as its primitive, from which subtraction, multiplication, and division are derived.

The project source files are organised as follows: `cpu_top.vhd` contains the top-level entity and all datapath logic; `alu.vhd` implements the FSM-controlled arithmetic unit; `binary_to_bcd.vhd` performs the double-dabble BCD conversion; and `seven_segment_decoder.vhd` provides the segment encoding. All files are synthesised together in a single Vivado project targeting the Artix-7 XC7A100T, with pin constraints defined in the accompanying `.xdc` constraints file.

4.3 Switch interface encoding

Mode	Switch	Function
Both	SW[8]	0 = Load mode, 1 = Execute mode
Load	SW[7]	Sign: 0 = positive, 1 = negative (two's complement)
Load	SW[3:0]	4-bit magnitude (0–15)
Load	SW[5:4]	Destination register R0–R3
Execute	SW[7:6]	Operation: 00=ADD, 01=SUB, 10=MUL, 11=DIV
Execute	SW[1:0]	Source register A (R0–R3)
Execute	SW[3:2]	Source register B (R0–R3)
Execute	SW[5:4]	Destination register R0–R3

4.4 PCB tools and FPGA

The custom PCB was designed in **Altium Designer** around the **Lattice iCE40UP5K-SG48I** FPGA. Power is supplied through two **AMS1117** LDO regulators generating VCC_3.3V and VCC_1.2V from a USB_5V input via a Mini-USB connector. The FPGA configuration bitstream is stored in a **Winbond W25Q16** SPI NOR Flash, connected via a four-wire SPI bus and clocked by a 12MHz oscillator (XLH735012).

The user interface consists of nine slide switches with 10 k Ω pull-up resistors, two push-buttons, a four-digit seven-segment display driven through NPN transistors (SBC817-40LT3G) for digit multiplexing, and eight status LEDs — all current-limited with 330 Ω resistors. The full schematic, covering all functional blocks, is shown in Figure 2.

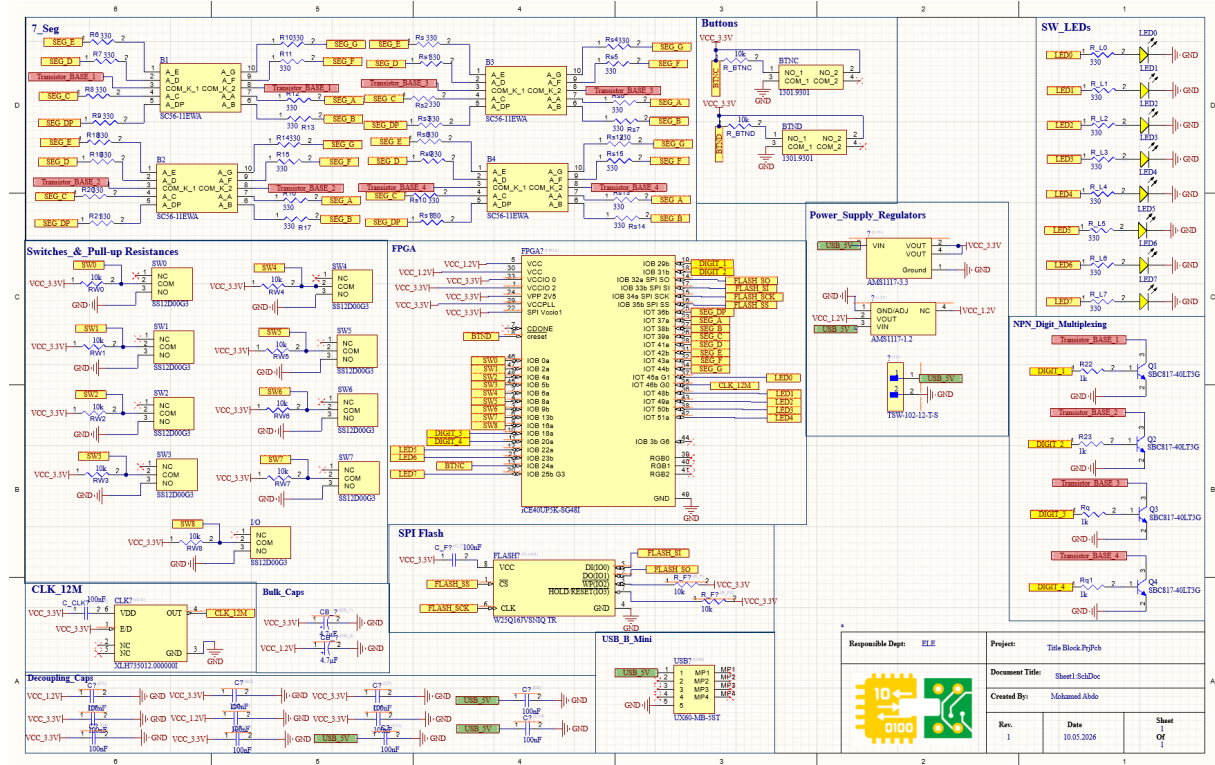


Figure 2: Full PCB schematic — all functional blocks: FPGA, power supply, SPI Flash, CLK, 7-segment display, switches, buttons, and LEDs

5 Implementation

5.1 CPU finite-state machine (FSM)

The CPU control logic is implemented as a **five-state** clocked FSM, with asynchronous reset on `btn_rst`. The states are: `S_IDLE`, `S_LOAD`, `S_START_ALU`, `S_WAIT_ALU`, and `S_WRITEBACK`.

- **S_IDLE** — the default state. The FSM waits for a rising edge on `btn_exec`. If `mode=LOAD`, it transitions to `S_LOAD`. If `mode=EXECUTE`, it selects the operation `op` $\in [0, 3]$ and transitions to `S_START_ALU`.
- **S_LOAD** — asserts `load_en = true`, writing the switch-encoded signed value into the selected destination register. Transitions back to `S_IDLE` immediately.
- **S_START_ALU** — latches operands A and B from the register file and the operation code into the ALU input registers. Pulses `alu_start` for one clock cycle. Transitions to `S_WAIT_ALU`.
- **S_WAIT_ALU** — holds until the ALU asserts `alu_done`. While `alu_done=0` the FSM stays in this state. Once `alu_done=1`, it transitions to `S_WRITEBACK`.
- **S_WRITEBACK** — asserts `writeback_en = true` and `store_flags = true`. Writes `alu_result` into `reg[dest]` and captures overflow, `div_zero`, and remainder flags. Returns to `S_IDLE`.

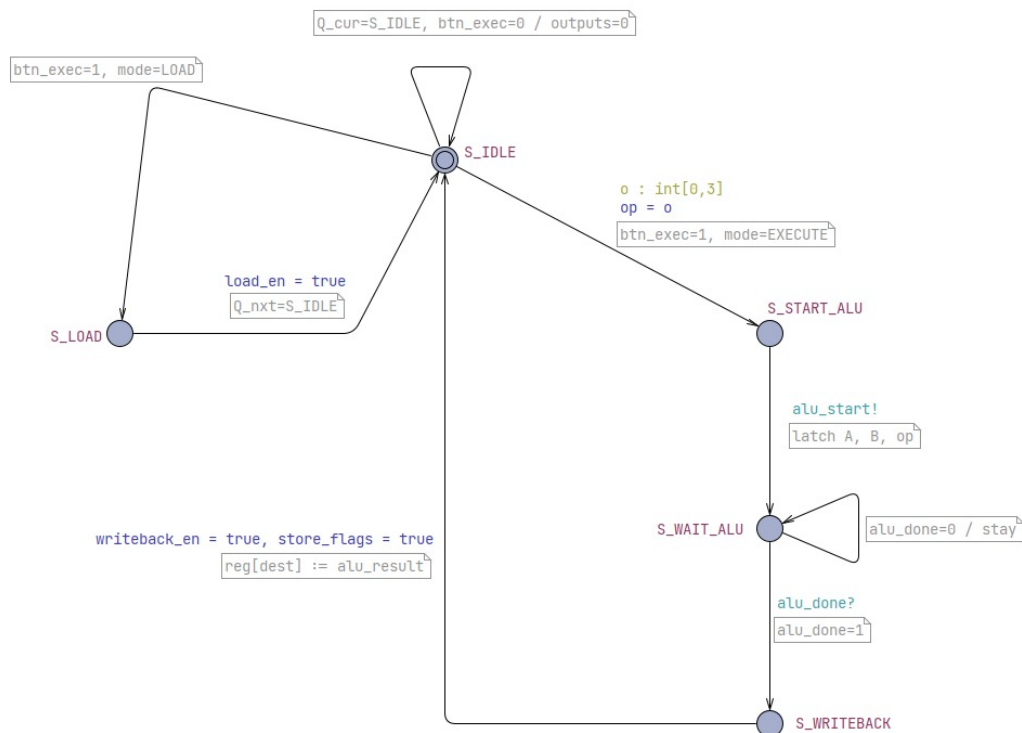


Figure 3: CPU FSM — five states covering idle, load, ALU start, wait, and writeback

5.2 ALU operations and flags

op	Operation	Notes
00	ADD	8-bit signed; carry and overflow flags set
01	SUB	Implemented as ADD with two's complement of B
10	MUL	8-bit $\times$ 8-bit; lower 8 bits stored as result
11	DIV	Quotient in result, remainder in <code>alu_remainder</code> ; <code>div_zero</code> on B=0

Two utility functions support arithmetic: `add8_ci` performs a ripple-carry 8-bit addition with carry-in, and `twos_comp8` computes the two's complement by inverting all bits and adding 1.

5.3 Display subsystem

The seven-segment display is driven by a **time-multiplexed refresh process** cycling through four digit positions at 50 kHz (refresh counter rolls at 49 999 clock cycles). Three display modes are handled:

- **Normal signed** — signed decimal with leading minus sign in the correct position (e.g. `-128` fills all four digits).
- **Division** (`stored_op=11`) — quotient with decimal point separating the remainder (e.g. `12.3` = quotient 12, remainder 3).
- **Error (OF)** — overflow or divide-by-zero displays `OF`.

5.4 PCB implementation — 4-layer stack-up

Table 1: PCB Layer Stack-Up

Layer	Function	Thickness	Purpose
L1	Top — Signals + Components	1.4 mil	FPGA, regulators, SPI Flash, UI
L2	GND Plane	1.378 mil	Solid copper, EMI shield, return path
L3	VCC_3.3 V Plane	1.378 mil	Power distribution, minimal routing
L4	Bottom — Signals + Caps	1.4 mil	Return signals, decoupling caps C1–C11

All vias are standard through-hole (Thru 1:4), spanning all four layers. Blind and buried vias were not required given the component density of this design and are not supported under JLCPCB’s standard 4-layer service.

5.5 Decoupling capacitor strategy

Table 2: Decoupling Capacitor Assignment

Reference	Value	Function
C1 – C8	100 nF	FPGA VCC_3.3 V & VCC_1.2 V decoupling
C9 – C11	100 nF	USB_5 V input line filtering
CB_1, CB_2	4.7 μ F	Bulk capacitors on 3.3 V and 1.2 V rails
C_F	100 nF	SPI Flash VCC bypass

All capacitors are placed **within 2 mm** of their associated FPGA power pin on the bottom layer, minimising parasitic inductance.

5.6 Trace widths and via rules

Table 3: Trace Width and Via Design Rules vs. JLCPCB Minimums

Net / Via	JLCPCB Min	Industry Std	Our Design
<i>Trace widths</i>			
GND	5 mil	15–25 mil	20 mil
USB 5 V	5 mil	20–30 mil	24 mil
VCC 3.3 V	5 mil	12–20 mil	16 mil
VCC 1.2 V	5 mil	10–15 mil	12 mil
Signals	5 mil	6–10 mil	8 mil
<i>Via dimensions (hole / pad)</i>			
Signal vias	0.3 / 0.5 mm	0.3 / 0.6 mm	12 / 24 mil
VCC 3.3 V	0.3 / 0.5 mm	0.4 / 0.8 mm	16 / 28 mil
Power GND	0.3 / 0.5 mm	0.45 / 0.9 mm	18 / 32 mil
USB 5 V	0.3 / 0.5 mm	0.45 / 0.9 mm	18 / 32 mil
Annular ring	0.1 mm	0.15 mm	6 mil

5.7 Routing and serpentine length tuning

The board was routed with the Situs autorouter (shortest-path, 6 mil clearance). Critical nets — CLK_12M and all three SPI Flash lines (FLASH_SCK, FLASH_SI, FLASH_SO) — were then **length-matched** using Altium's Interactive Length Tuning (Shift+A) to eliminate propagation skew across the parallel bus. Settings: amplitude 0.3–0.5 mm, gap 0.15 mm, rounded corners. The result was **0 DRC violations** after tuning.

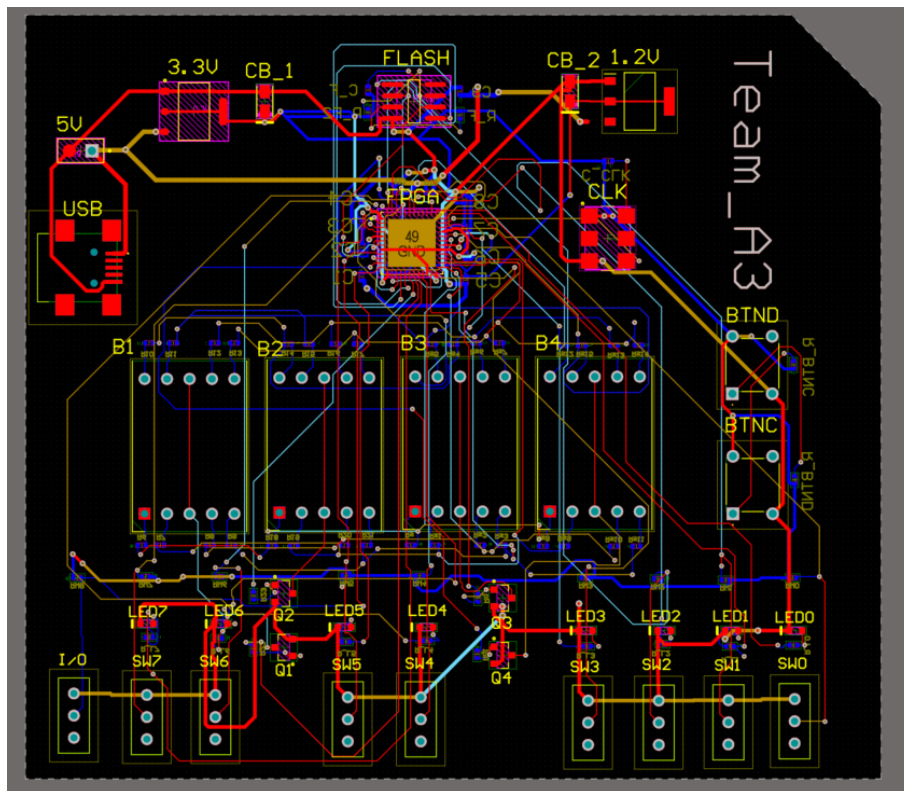


Figure 4: PCB layout — serpentine length tuning on CLK_12M and SPI Flash lines

6 Results

6.1 VHDL simulation and on-board testing

The complete CPU was simulated in Vivado before synthesis. All four ALU operations were verified with both positive and negative operands. Edge cases tested include:

- Signed overflow: e.g. $127 + 1$ triggers the **overflow** flag and displays 0F.
- Division by zero: $B=0$ triggers **div_zero**, display shows 0F.
- Division display: e.g. $25 \div 4$ shows 6.1 (quotient 6, remainder 1) with the decimal point active.
- Negative display: e.g. -128 correctly drives all four digit positions including the leading minus.

After synthesis and programming, all cases were confirmed on the Nexys A7 board.

6.2 Vivado synthesis results

After synthesis and implementation, Vivado reported the following resource utilisation on the Artix-7 XC7A100T. The design uses a small fraction of the available device resources, confirming that the 8-bit CPU fits comfortably within the target FPGA.

Name	Slice LUTs (63400)	Slice Registers (126800)	Slice (15850)	LUT as Logic (63400)	Bonded IOB (210)	BUFGCTRL (32)
cpu_top	1361	197	420	1361	36	1
ALU_UNIT (alu)	167	110	55	167	0	0
DIV_UNIT (divider)	75	48	26	75	0	0
MULT (multiplier)	60	43	24	60	0	0
BCD_NORMAL (binary_to_bcd)	611	0	193	611	0	0
BCD_REM (binary_to_bcd_0)	533	0	168	533	0	0

Figure 5: Resource utilisation report — Vivado implementation

The total design consumes 1361 Slice LUTs (2.1% of 63400) and 197 Slice Registers (0.16% of 126800), leaving the vast majority of the device available for future logic expansion. The BCD conversion modules (BCD_NORMAL and BCD_REM) account for the largest share of LUT usage (611 and 533 respectively), as the double-dabble algorithm is purely combinational and maps entirely to look-up tables with no registers. The ALU including the divider and multiplier consumes only 167 LUTs and 110 registers, reflecting the efficiency of the sequential FSM-based approach. Only 36 of the 210 available bonded IOBs are used, consistent with the limited number of switches, buttons, LEDs, and display signals on the Nexys A7. The netlist floorplan confirms that all logic is concentrated in a single tile region (X0Y1), with the rest of the device completely unused.

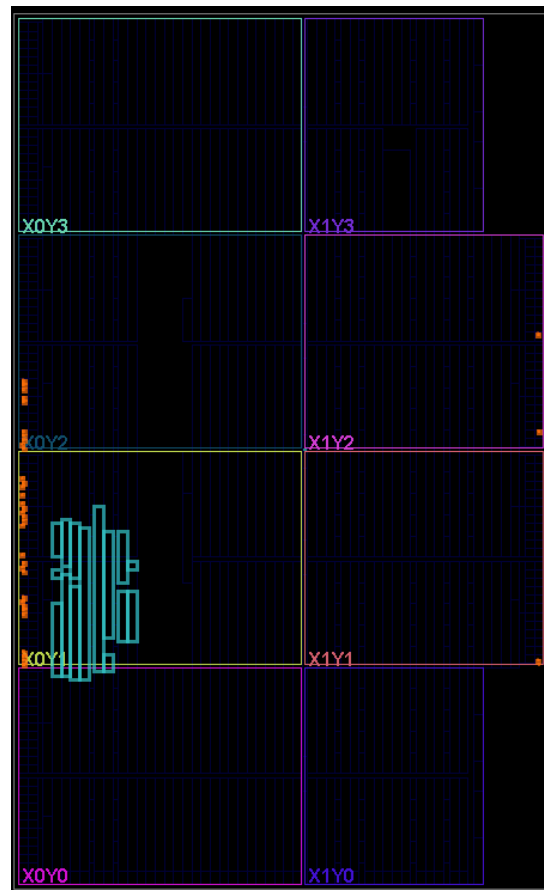


Figure 6: Vivado netlist floorplan — logic concentrated in tile X0Y1

6.3 PCB manufacturing compliance

Table 4: JLCPCB Manufacturing Compliance

Parameter	JLCPCB Min	Industry Std	Our Design
Min trace width	5 mil (0.127 mm)	6–8 mil	8 mil
Min clearance	5 mil (0.127 mm)	6 mil	6 mil
Via hole (signal)	0.3 mm (11.8 mil)	0.3 mm	12 mil (0.305 mm)
Via pad (signal)	0.5 mm (19.7 mil)	0.6 mm	24 mil (0.61 mm)
Annular ring	0.1 mm (3.94 mil)	0.15 mm	6 mil (0.15 mm)
Power trace (3.3 V)	5 mil min	12–20 mil	16 mil
Power trace (GND)	5 mil min	15–25 mil	20 mil
Hole-to-hole spacing	0.5 mm edge	0.5 mm	Verified via DRC
Layer count	Standard 4-layer	4-layer	4 layers

6.4 DRC result — 0 violations

DRC Result: 0 Violations

All five DRC categories confirmed clear:

- Clearance: 0 violations
- Short-circuit: 0 violations
- Un-routed nets: 0 violations
- Net antennae: 0 violations
- Hole-to-hole: 0 violations

The Gerber files are ready for production submission to JLCPCB.

Sources / References

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